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2 / 2 submitted

Q1. Smart Hospital Patient Registration System

Score: 0.00 TC: 0/3 passed

CODE

```
import java.util.*;
```

INPUT

3
101
Dr.Arun
CSE
60000
102
Mr.Ravi
ECE
39500
103
Dr.Kumar
IT
48000
102

OUTPUT

Employee Details

Employee ID : 101
Name : Dr.Arun
Department : CSE
Monthly Salary : 60000.0

Employee ID : 102
Name : Mr.Ravi
Department : ECE
Monthly Salary : 39500.0

Employee ID : 103
Name : Dr.Kumar
Department : IT
Monthly Salary : 48000.0

Highest Paid Employee
Employee ID : 101
Salary : 60000.0

Average Salary
49166.67

Employee Found
Employee ID : 102
Name : Mr.Ravi
Salary : 39500.0

```
        for (String a : authors) {
            System.out.println(a);
        }
    } else if (choice == 5) {
        HashMap<String, Integer> catCount = new HashMap<>();

        for (int i = 0; i < books.size(); i++) {
            Book b = books.get(i);
            String c = b.category;

            Integer old = catCount.get(c);
            if (old == null) catCount.put(c, 1);
            else catCount.put(c, old + 1);
        }

        System.out.println("Category-wise Count");
        for (String key : catCount.keySet()) {
            System.out.println(key + " : " + catCount.get(key));
        }

        for (int i = 0; i < books.size(); i++) {
            Book b = books.get(i);
            System.out.println("Book Details");
            System.out.println("Book ID : " + b.id);
            System.out.println("Title : " + b.title);
            System.out.println("Author : " + b.author);
            System.out.println("Category : " + b.category);
            System.out.println("Publisher : " + b.publisher);

            String t = b.title.trim();
            int len = b.title.length();

            int words = 0;
            if (t.length() > 0) {
                words = 1;
                for (int k = 0; k < t.length(); k++) {
                    if (t.charAt(k) == ' ') words++;
                }
            }

            System.out.println("Length : " + len);
            System.out.println("Words : " + words);
            System.out.println("Category : " + b.category);
        }
    }
}

sc.close();
}
```

INPUT

```
1
B101
Herbert Schildt
PROGRAMMING
McGraw Hill
2
B101
3
4
5
6
```

OUTPUT

(no output)

```

while (true) {
    if (!sc.hasNextLine()) break;
    String line = sc.nextLine().trim();
    if (line.length() == 0) continue;

    int choice;
    try {
        choice = Integer.parseInt(line);
    } catch (Exception e) {
        continue;
    }

    if (choice == 6) break;

    if (choice == 1) {
        if (books.size() >= maxBooks) continue;

        String id = trimSafe(sc.nextLine());
        String title = trimSafe(sc.nextLine());
        String author = trimSafe(sc.nextLine());
        String category = trimSafe(sc.nextLine());
        String publisher = trimSafe(sc.nextLine());

        boolean ok = true;

        if (id.length() == 0) ok = false;
        if (title.length() == 0) ok = false;
        if (author.length() == 0) ok = false;
        if (category.length() == 0) ok = false;
        if (publisher.length() == 0) ok = false;
        if (bookById.containsKey(id)) ok = false;

        if (!ok) continue;

        title = toTitleCase(title);
        category = category.toUpperCase();

        Book b = new Book(id, title, author, category, publisher);
        books.add(b);
        bookById.put(id, b);
        authors.add(author);

        System.out.println("Book Added Successfully.");
    } else if (choice == 2) {
        String id = trimSafe(sc.nextLine());
        Book b = bookById.get(id);
        if (b == null) {
            System.out.println("Book Not Available");
        } else {
            System.out.println("Book ID : " + b.id);
            System.out.println("Book Title : " + b.title);
            System.out.println("Author : " + b.author);
            System.out.println("Category : " + b.category);
            System.out.println("Publisher : " + b.publisher);
        }
    } else if (choice == 3) {
        System.out.println("-----");
        System.out.println("ID TITLE AUTHOR CATEGORY");
        System.out.println("-----");
        for (int i = 0; i < books.size(); i++) {
            Book b = books.get(i);
            System.out.println(b.id + " " + b.title + " " + b.author + " " + b.category);
        }
    } else if (choice == 4) {
        System.out.println("Unique Authors");
    }
}

```

Q2. Online Library Book Management System**CODE**

```
import java.util.Scanner;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.HashSet;

public class Main {

    static class Book {
        String id;
        String title;
        String author;
        String category;
        String publisher;

        Book(String id, String title, String author, String category, String publisher) {
            this.id = id;
            this.title = title;
            this.author = author;
            this.category = category;
            this.publisher = publisher;
        }
    }

    static String trimSafe(String s) {
        if (s == null) return "";
        return s.trim();
    }

    static String toTitleCase(String s) {
        s = trimSafe(s);
        if (s.length() == 0) return s;

        s = s.toLowerCase();
        StringBuilder out = new StringBuilder();
        boolean newWord = true;

        for (int i = 0; i < s.length(); i++) {
            char ch = s.charAt(i);
            if (ch == ' ') {
                out.append(' ');
                newWord = true;
            } else {
                if (newWord) {
                    if (ch >= 'a' && ch <= 'z') ch = (char) (ch - 'a' + 'A');
                    newWord = false;
                }
                out.append(ch);
            }
        }

        return out.toString().replaceAll(" +", " ");
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        ArrayList<Book> books = new ArrayList<>();
        HashMap<String, Book> bookById = new HashMap<>();
        HashSet<String> authors = new HashSet<>();

        int maxBooks = 500;
```